

$$\sqrt{\langle c_f^2 \rangle} = \sqrt{2} \sqrt{\langle c^2 \rangle}$$

$$= 1.4 \sqrt{\langle c^2 \rangle}$$

15

D

Amplitude is constant for a particle in simple harmonic motion; it is displacement that changes.

Angular frequency = $2\pi f$ and is constant.

Total energy is constant as there is no dissipative force present.